

# FuturePath Eswatini

## Platform Documentation & Partnership Proposal

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Submitted to: Likusasa Letfu

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## 1. Executive Summary

FuturePath Eswatini is a fully functional, production-ready digital platform built to connect Emaswati youth with employment opportunities, internships, professional development tools, and structured online learning. It was designed and developed entirely by Mnelisi Simelane, a 20-year-old self-taught software engineer from Mbabane, Eswatini, with product design and quality review contributions from Thandolwethu Lukhele.

The platform is not a concept, mockup, or prototype. It is a complete, deployed web application — built using industry-standard technologies — that is ready to serve real users from day one.

This document presents FuturePath Eswatini to Likusasa Letfu with a formal proposal: the full platform is offered as a donation to Likusasa Letfu, along with hands-on pilot support, staff onboarding, and ongoing technical assistance. In return, the builders respectfully request scholarship support for their respective university studies abroad.

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## 2. Background & The Problem

### 2.1 The Challenge Facing Emaswati Youth

Youth unemployment is one of the most significant challenges facing the Kingdom of Eswatini. Despite a growing number of young people completing secondary and tertiary education, a large proportion remain unable to access structured career pathways, quality learning resources, or professional tools that can give them a competitive edge in the job market.

Key problems identified include:

- No single centralised digital platform connecting Emaswati youth to local employers and internship opportunities.
- Limited access to career guidance, professional CV tools, and interview preparation resources.
- A gap between the skills employers expect and those young people currently demonstrate on applications.
- No digital certificate system allowing youth to verify and share qualifications earned through short courses.
- A lack of locally-contextualised learning content tailored to the Eswatini corporate and economic landscape.

## **2.2 Why FuturePath Was Built**

FuturePath Eswatini was built from the ground up to address these gaps — not as a foreign solution imported and adapted, but as something designed with deep knowledge of the local context: the employers, the culture, the economic realities, and the specific challenges young Emaswati face when entering the workforce.

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# **3. Platform Overview**

## **3.1 What is FuturePath Eswatini?**

FuturePath Eswatini is a responsive web application accessible from any device — desktop, tablet, or mobile — that provides Emaswati youth with a single platform for career development, job seeking, professional learning, and credential verification.

### 3.2 Tech Stack

| Layer              | Technology Used                                      |
|--------------------|------------------------------------------------------|
| Frontend Framework | React 18 + TypeScript                                |
| Styling            | Tailwind CSS + custom design system                  |
| Routing            | React Router v6                                      |
| Backend & Database | Supabase (PostgreSQL)                                |
| Authentication     | Supabase Auth (email/password + session persistence) |
| PDF Generation     | jsPDF (CVs and Certificates)                         |
| QR Codes           | qrcode library                                       |
| State Management   | Zustand + React Hook Form                            |
| Hosting            | Vercel / Netlify (deployable immediately)            |

### 3.3 Database Tables

The platform uses a fully designed PostgreSQL database with 14 tables, complete relationships, indexes, and Row Level Security (RLS) policies:

| Table        | Purpose                                                       |
|--------------|---------------------------------------------------------------|
| profiles     | Extended user data — name, phone, role (user/admin)           |
| jobs         | All job listings with type, category, salary, requirements    |
| internships  | Internship listings from NGOs, government, and private sector |
| applications | User applications for both jobs and internships               |

|                |                                                            |
|----------------|------------------------------------------------------------|
| saved_jobs     | Jobs bookmarked by users                                   |
| courses        | Learning courses with category, level, and duration        |
| course_modules | Individual lessons and video modules within each course    |
| quizzes        | Quizzes linked to course modules                           |
| quiz_questions | Questions, multiple-choice options, and correct answers    |
| quiz_attempts  | User quiz submissions, scores, and pass/fail records       |
| certificates   | Issued certificates with unique IDs and revocation support |
| notifications  | In-app notifications for jobs, courses, and certificates   |
| user_progress  | Per-course completion percentage, streaks, and hours       |
| cv_data        | JSONB storage of each user's full CV content               |

## 4. Platform Features

### 4.1 Authentication & User Accounts

- Email and password registration with full name and phone number
- Email verification flow and password reset via email
- Session persistence — users stay logged in across sessions
- Role-based access: standard user and admin

### 4.2 User Dashboard

- Personalised welcome with the user's name and avatar
- Active course progress ring showing completion percentage
- Career path timeline — showing verified degree, current course, and next milestone
- Recommended jobs carousel populated from the live job database
- Quick-access stats: saved jobs, applications submitted, courses completed, certificates earned
- Build CV prompt and verification status indicator

### 4.3 Jobs Module

- Searchable listings from top Eswatini employers — MTN Eswatini, Standard Bank, UNICEF, Nedbank, Eswatini Water Services, Deloitte, and more
- Filter by employment type: Full-time, Internship, Remote, Part-time, Contract
- Filter by category: Technology, Finance, Business, Engineering, Education, Healthcare, Marketing, Government
- Each listing includes: job title, company, location, salary, experience level, full description, and requirements
- Apply directly in-platform — applications stored in the database with status tracking
- Save / unsave jobs for later review
- Application Sent confirmation screen with next-steps timeline

### 4.4 CV Builder

- Guided 4-step wizard: Personal Information → Education → Work Experience → Skills
- Dynamic field arrays — add and remove multiple education entries, work roles, certifications, and projects
- Skill tag manager with suggested Eswatini-relevant skills
- ATS-optimised live preview of the formatted CV
- One-click PDF export using jsPDF — professionally formatted and downloadable
- Auto-save to the user's Supabase record on each step

- Sections: Personal Info, Education, Experience, Skills, Certifications, Projects, References

#### 4.5 Learning Academy

- Courses organised by category: Technology, Business, Finance, Practical Skills, Arts & Creativity
- Each course includes: video module player, lesson summaries, downloadable resources, and a quiz
- Course progress tracking with visual progress bars per course
- Module progress tracker showing completed, current, and locked modules
- Search and filter courses by name and category

#### 4.6 Quiz Engine

- Multiple-choice question system with 4 options per question
- Instant grading — correct answers highlighted in green, wrong in red
- Pass/fail scoring — 60% pass mark by default (configurable per quiz)
- Full retake support — unlimited attempts
- Quiz progress bar and question counter during the attempt
- Results screen with score percentage and next-step options

#### 4.7 Certificate System

- Certificates issued automatically upon course completion
- Each certificate includes: user name, course name, date issued, unique certificate ID, and QR verification code
- Certificates stored in the database and viewable in the user's Verify tab
- PDF download of the full certificate (landscape A4, professionally designed)
- Share link — copy a verification URL to clipboard
- Public verification portal — anyone can enter a certificate ID to confirm its authenticity

- Admin can revoke certificates if required

#### 4.8 Interview Preparation Module

- Eswatini-specific Corporate Q&A — common interview questions with locally-contextualised model answers
- Technical readiness checklist for sector-specific interviews
- Local etiquette and cultural guidance for Eswatini corporate environments
- AI-assisted mock interview session with real video-feed UI, live feedback on tone and pacing, audio level indicator, and question navigation
- Resource library: interview prep guides and downloadable PDF resources

#### 4.9 Notifications

- In-app notification bell with unread badge count
- Notification types: new jobs, new internships, course completion, new courses, certificate issued
- Slide-up notification panel with mark-all-as-read functionality

#### 4.10 Admin Dashboard

The admin dashboard is accessible to authorised personnel only and provides full management control over the platform:

- Jobs: Add, edit, and delete job listings with a full form including type, category, salary, description, and requirements
- Courses: Add and delete courses with category, level, duration, and description
- Users: View all registered users, suspend accounts, and manage roles
- Certificates: View all issued certificates and revoke them if required
- Overview dashboard with live stats: total users, active jobs, total courses, and certificates issued
- Recent activity feed showing the latest registrations, applications, and certificate issuances

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## 5. The Partnership Proposal

### 5.1 What Likusasa Letfu Receives

Mnelisi Simelane and Thandolwethu Lukhele are offering to donate FuturePath Eswatini in its entirety to Likusasa Letfu, and to personally support the organisation in piloting, customising, and launching the platform for the youth it serves. The full offering includes:

1. Full handover of the complete source code, database schema, and all deployment configuration files.
2. Deployment of the live platform on Vercel or Netlify, both of which offer free hosting tiers sufficient for the pilot phase.
3. Guided staff onboarding — hands-on training sessions for Likusasa Letfu personnel on using the admin dashboard, managing job listings, publishing courses, and verifying certificates.
4. Customisation of the platform to reflect Likusasa Letfu's brand identity, colour scheme, and programme structure.
5. A structured pilot and launch plan, developed collaboratively with Likusasa Letfu, targeting youth in the organisation's existing beneficiary network.
6. Ongoing remote technical support and maintenance assistance during the pilot phase, accessible via email and messaging.

### 5.2 The Request — University Scholarships

In exchange for the full donation and support described above, Mnelisi Simelane and Thandolwethu Lukhele respectfully request that Likusasa Letfu consider providing university scholarship support for both team members:

**Mnelisi Simelane** — Software Engineering · China · Commencing 2026

Requesting tuition fee scholarship to support a degree in Software Engineering at a university in China.

**Thandolwethu Lukhele** — BSc Business Management · Botswana · Commencing 2026

Requesting tuition fee scholarship to support a BSc in Business Management at a university in Botswana.

Both students are committed to completing their studies and returning to Eswatini with internationally recognised qualifications to continue contributing to the country's digital and business development.

### 5.3 Preparedness Assistance

In addition to tuition fee scholarships, the team respectfully requests preparedness assistance to cover the practical costs of preparing to study abroad — including travel documentation, initial living costs, and relocation preparation. This assistance is requested for both Mnelisi Simelane and Thandolwethu Lukhele.

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## 6. About the Team

### 6.1 Mnelisi Simelane — Lead Developer

**Age:** 20 years old

**Location:** Mbabane, Eswatini

**Contact:** mnethedev@gmail.com

**Role:** Full-stack Software Engineer — Design, Development & Deployment

Mnelisi is a self-taught software engineer who has spent several years building real, production-deployed applications targeting the Eswatini market. He has no formal degree — every skill has been built through deliberate practice, self-study, and real-world application.

Projects built by Mnelisi Simelane:

- FuturePath Eswatini — Youth career and employment platform (this document)
- EduLink Eswatini — National high school admissions platform

- SwiftWallet — Digital payments MVP for the Eswatini market
- TM HealthOS — Connected healthcare infrastructure platform
- Stoic Tunes — Music project platform (DJ and producer: Deep House, Afro House, Afro Tech)

## 6.2 Thandolwethu Lukhele — Contributor

**Role:** Product Design & Quality Review

Thandolwethu Lukhele contributed to FuturePath Eswatini through product design input and quality review, ensuring the platform meets the usability and experience standards required to serve Emaswati youth effectively.

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# 7. Proposed Pilot Plan

## Phase 1 — Setup & Customisation (Week 1–2)

- Deploy the platform to a live domain
- Customise branding to reflect Likusasa Letfu identity
- Populate initial job listings and courses relevant to the beneficiary base
- Train admin staff on the dashboard

## Phase 2 — Soft Launch (Week 3–4)

- Onboard a pilot group of 20–50 youth from the existing Likusasa Letfu network
- Collect feedback on registration, job browsing, CV building, and learning modules
- Monitor application activity and course engagement

## Phase 3 — Full Launch (Month 2+)

- Open registration to all Emaswati youth

- Add additional employer partnerships and course content
- Issue first certificates to course completers
- Review platform performance and plan feature improvements

Mnelisi Simelane will provide remote technical support throughout all three phases and remain available for troubleshooting, feature updates, and guidance while studying abroad.

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## 8. Deployment & Technical Requirements

### 8.1 Hosting

The platform is deployable immediately on Vercel or Netlify — both offer free hosting tiers that are sufficient for the pilot phase at no cost to Likusasa Letfu. Scaling to a paid plan would only be required once user numbers grow significantly.

### 8.2 Database

The database runs on Supabase, which provides a free PostgreSQL tier sufficient for thousands of users. The complete SQL schema is included in the codebase handover — all tables, indexes, Row Level Security policies, and seed data are ready to deploy in a single step.

### 8.3 Environment Requirements

To deploy the platform, the following are required:

- A Supabase account (free) — [supabase.com](https://supabase.com)
- A Vercel or Netlify account (free) — [vercel.com](https://vercel.com) or [netlify.com](https://netlify.com)
- A domain name (optional for pilot — a free subdomain is provided by both hosts)
- Two environment variables: `VITE_SUPABASE_URL` and `VITE_SUPABASE_ANON_KEY` (provided in the handover)

## 8.4 Ongoing Costs

| Service                      | Cost (Pilot Phase)                       |
|------------------------------|------------------------------------------|
| Supabase (Database & Auth)   | Free — up to 50,000 users                |
| Vercel / Netlify (Hosting)   | Free — unlimited deployments             |
| Domain name (optional)       | ~\$12/year if a custom domain is desired |
| Total estimated monthly cost | E0 — E0 for the pilot phase              |

## 9. Closing Statement

FuturePath Eswatini represents something rare: a complete, working digital platform built locally, by young Emaswati, for young Emaswati — at no cost to the organisation that would deploy it. Every line of code, every design decision, and every feature was shaped by a genuine understanding of what our youth need to succeed in the Eswatini job market.

We are not asking for funding to build something. We are offering something already built, and asking only for the opportunity to continue growing — so that we can return to Eswatini better equipped to keep building.

We would be honoured to present FuturePath to the Likusasa Letfu team in person, walk through a live demo of every feature, and discuss how a partnership might be structured in a way that works best for your organisation and the youth you serve.

Thank you sincerely for your time and consideration.

**Mnelisi Simelane & Thandolwethu Lukhele**

FuturePath Eswatini · mnethedev@gmail.com · Mbabane, Eswatini